

FSU SAC AP Poll Model: R Files and what they do

File	Description
APPollDataCollection.R	• Loads each teams schedule from 2014-2024 • Joins schedule data with poll data on team, season, week • Joins a df called “next_week” with poll data to determine how many AP Poll points were gained/lost from one week to the next • Gets the number of poll points that a team’s opponent has • Fits a simple linear regression model (not the final model!) • Creates no_byes, which is a dataframe of all team-weeks where the team did not have a bye (not used in final model) • Writes it to a csv called APPollData.csv (currently commented out)
APPollScraping.R	• Creates a function (get_ap_voters) that scrapes collegepolltracker.com (CPT) that, for a given year and week, returns each voter and their top 25 (not used in the model afaik) • Creates a function (get_ap_poll) that scrapes CPT for a given year and week, returns all the teams that got AP votes and how many • Loops get_ap_poll over 2014-2024 weeks 3-15 to get all of those polls
APPollCleaningTest.R	• Removes a useless column from APPollData.csv • Converts NAs for Opponent Poll Points to 0 • Removes 2020 data • Adds the Rank for each team in that week • Adds the next week that team was ranked

	• Adds the rank of the team next week • Converts game result to a binary (1 if W, 0 if L) • Adds bias for the teams determined in R&D meeting • Some linear regression models that don't mean anything • Creates poll_data_summary that keeps only the team, year, week, rank, next_week_rank vs. what LR model predicted
APPollCleaning.R	• Same as APPollCleaningTest.R, just without the converting W/L to 1/0
APPollWithRecordsOLD.R	• Uses CFBFastR's cfb_game_info function to get game data from 2014-2024 • Creates team_games, which has just each teams games played and one-hot encoded win/loss • Creates weekly_records, a running tally of each teams recorded (with columns total_wins, total_losses and record, which is total_wins '-' total_losses) for all years outside of 2020 • Gets max_poll_weeks, the last week in which we have poll data for each year • Creates weekly_records_full, which fills in team's records during bye weeks • Fixes an issue with Miami (FL) being called Miami • Joins records with existing poll_data • Tests teams that don't have a record for whatever reason (shouldn't be needed anymore)
APPollCleaning2.R	• Removes a useless column from APPollData.csv • Converts NAs for opponent Poll Points to 0s • Removes 2020 data • Adds the Rank for each team in that week

	• Adds the next week that team was ranked • Adds the rank of the team next week • Converts game result to a binary (1 if W, 0 if L) • Adds bias for the teams determined in R&D meeting (note: these biases look more accurate than APPollCleaning.R, so I most likely used this one) • Some linear regression models that don't mean anything • Creates poll_data_summary that keeps only the team, year, week, rank, next_week_rank vs. what LR model predicted • Writes poll_data to CFBPollDataFeb16.csv
APPollWithRecords.R	• Fixes the Miami/Miami (FL) issue • Uses CFBFastR's cfb_game_info function to get game data from 2014-2024 • Uses above to create two dfs, one for home games and one for away games, with conference data and one-hot encoded win/loss, then merges home and away • Creates weekly_records, a running tally of each teams recorded (with columns total_wins, total_losses and record, which is total_wins '-' total_losses) for all years outside of 2020 • Gets max_poll_weeks, the last week in which we have poll data for each year • Creates weekly_records_full, which fills in team's records during bye weeks • Added opponent_lookup, which gets the record and conference for a team's opponent in that week, and joins it with weekly_records_full • Writes this to CFBTeamResults.csv

	• Joins this data with poll_data (from Feb16.csv) on team, year, week and sorts by rank
APPollDataCleaningMess.R	• Uses CFBPollDataFeb23.csv for poll data and CFBTeamResults for game results • ChatGPT function to calculate win streak, don't believe it was the final one we ended up using • Adds it to team_records • Another (failed) attempts at calculating win streak
APPollWithRecordsCLEAN.R	• Uses Feb16.csv • Fixes the Miami/Miami (FL) issue • Uses CFBFastR's cfbd_game_info function to get game data from 2014-2024 • Uses above to create two dfs, one for home games and one for away games, with conference data and one-hot encoded win/loss, then merges home and away • Creates team_records, a running tally of each teams recorded (with columns total_wins, total_losses and record, which is total_wins '-' total_losses) for all years outside of 2020 • Gets max_week, the last week in which we have poll data for each year • Creates weekly_records_full, which fills in team's records during bye weeks • Added opponent_lookup, which gets the record and conference for a team's opponent in that week, and joins it with weekly_records_full • Merges everything into final data, selects only the relevant columns, renames columns • Writes weekly_records full to CFBTeamResults.csv • Writes final_data to CFBPollDataFeb23.csv

APPollAddingStreak.R	• Uses CFBTeamResultsPRE.csv for records • Adds an undefeated column • Adds a win streak function to the df • Gets poll data from Feb23.csv • Adds streak to final_data • Separates wins/losses to 2 variables each, one for total wins/losses before that week's game and one for total wins/losses after that week's game (if there was one) • Drops a useless column • Writes this to CFBPollDataMar2.csv • Adds opponent wins/losses entering the game • Writes this to CFBPollDataFeb23.csv (???) • Creates some test dfs (ignore)
APPollWithRecords3.R	• Uses Feb16.csv • Fixes the Miami/Miami (FL) issue • Uses CFBFastR's cfb_d_game_info function to get game data from 2014-2024 • Uses above to create two dfs, one for home games and one for away games, with conference data and one-hot encoded win/loss, then merges home and away • Creates team_games, a running tally of each teams recorded (with columns total_wins, total_losses and record, which is total_wins '-' total_losses) for all years outside of 2020 • Gets max_week, the last week in which we have poll data for each year • Creates weekly_full, which fills in team's records during bye weeks • Added opponent_lookup, which gets the record and conference for a team's opponent in that week, and joins it with weekly_full • Merges everything into final data, selects only the relevant columns, renames columns • Adds column undefeated teams

	• Fills in NAs for columns w/ poll points with 0 • Renames columns that got messed up during a join • Writes weekly full to CFBTeamResults.csv • Adds bias to the df (do this before the write.csv line) if not already there • Writes final_data to CFBPollDataFinal.csv
APPollBasicMLR.R	• Filters out Miami (somehow all the times I “fixed” it didn’t work) and week 15 • Creates a set of predictor variables, encodes the categorical variables (result, conferences) as factors • Runs linear regression on all the predictors, plots how well the model did • Calculates the error for each team, creates a df (farthest_off) that lists the 10 biggest misses • Creates avg_points_by_rank, which lists how many points, on average, each rank has • Gets the RMSE of the LR model
APPollFillingNAs.R	• Uses CFBPollDataFinal.csv • Fixes an issue where San Jose State did not have a conference listed for their games • Replaced all unknown opponent conferences with a blanket “FCS” (they aren’t all FCS, some are lower division, but it was the quick and easy fix, most of these teams aren’t ranked anyway) •